



AIMS

African Institute for
Mathematical Sciences
RWANDA

Introduction to Probability and Statistics

Lecture 11: Confidence Intervals

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1 Useful results

2 Confidence Intervals

Combinations of r.v.'s

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Chi-square distribution

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The summation of the square of n *independent standard normal r.v.'s* follows a **Chi-square** distribution with n degrees of freedom (d.o.f.).

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A r.v. X has a **chi-square** distribution with n degrees of freedom (d.o.f.) iff its p.d.f has the form:

$$f_X(x) = \frac{2^{-n/2}}{\Gamma(n/2)} x^{n/2-1} e^{-x/2}, \quad x \in \mathbb{R}^+, n \in \mathbb{N}^*.$$

Notation: $X \sim \chi^2(n)$. The c.d.f. is denoted: $\chi_n^2(x) = P(X \leq x)$.

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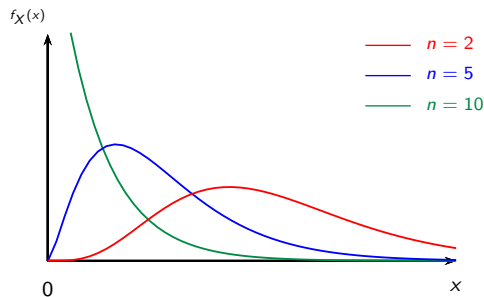
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The p.d.f. of a **Chi-square** r.v. with n d.o.f. is the p.d.f of a **gamma distribution** with parameters $(n/2, 1/2)$.

Chi-square distribution

P.d.f of a Chi-2 r.v for different d.o.f.



Combinations of r.v.'s

● **Student's t-distribution**

- Let $Z \sim \mathcal{N}(0, 1)$ and $Y \sim \chi^2(n)$ be two independent r.v.'s, then:

$$T = \frac{Z}{\sqrt{Y/n}}$$

follows a **Student's t-distribution** with n degrees of freedom.

- Notation: $Z \sim \mathcal{T}(n)$.
- The p.d.f. of a **Student's t-distribution** is:

$$f_T(t) = \frac{\Gamma[(n+1)/2]}{\sqrt{n\pi} \Gamma(n/2)} \cdot \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2}, \quad t \in \mathbb{R}^+.$$

- The c.d.f. is denoted: $t_n(x)$.

Combinations of r.v.'s

● **Fisher distribution**

- Let $Y_1 \sim \chi^2(n_1)$ and $Y_2 \sim \chi^2(n_2)$, be two independent r.v.'s, then

$$Y = \frac{Y_1/n_1}{Y_2/n_2}$$

follows a **Fisher distribution** with (n_1, n_2) degrees of freedom.

- Notation: $Y \sim \mathcal{F}(n_1, n_2)$.
- The p.d.f. of a **Fisher distribution** with (n_1, n_2) degrees of freedom is:

$$f_Y(y) = \frac{\Gamma[(n_1 + n_2)/2]}{\Gamma(n_1/2) \Gamma(n_2/2)} \cdot \left(\frac{n_1}{n_2}\right)^{n_1/2} \cdot \frac{y^{(n_1-2)/2}}{(1 + n_1 y/n_2)^{(n_1+n_2)/2}}.$$

Joint moments generating function

Definition

Let $X_1, X_2, \dots, X_n$ be n r.v.'s. The joint moments generating function of $(X_1, \dots, X_n)$ is:

$$M_{X_1, \dots, X_n}(t_1, \dots, t_n) = E\left(e^{t_1 X_1 + \dots + t_n X_n}\right),$$

where $t_1, \dots, t_n$ are real numbers.

Proposition

Let X_1, X_2 be two r.v.'s. X_1 and X_2 are independent iff the joint moments generating function of X_1, X_2 is the production of m.g.f. of X_1 and X_2 .

$$M_{X_1, X_2}(t_1, t_2) = M_{X_1}(t_1) M_{X_2}(t_2), \quad \forall t_1, t_2.$$

Confidence Intervals

Definition

A $(1 - \alpha)$ -confidence interval (C.I.) for a parameter θ is an interval in $\Theta \subseteq \mathbb{R}$ that has a probability $1 - \alpha$ to contain the true value of θ .

- Constructing a $(1 - \alpha)$ -confidence interval is finding an interval (a, b) such that $P((a, b) \ni \theta) = 1 - \alpha$ that is

$$P(a < \theta \leq b) = 1 - \alpha.$$

- The methods will rely on the properties of the estimators.

First strategy: pivotal function

Definition

A pivotal function is a statistic depending on the parameter of interest such that its probability distribution does not depend on the unknown parameters.

Example:

Suppose that a sample $X_1, \dots, X_n$ of n independent r.v.'s from an exponential distribution with parameter θ .

To build a C.I. for θ , we consider the pivotal function: $W = \frac{2n\bar{X}_n}{\theta}$.

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Possible bounds: $a = \frac{2n\bar{X}_n}{\chi_{2n}^{2-1}(1-\alpha/2)}$ and $b = \frac{2n\bar{X}_n}{\chi_{2n}^{2-1}(\alpha/2)}$.

First strategy: pivotal function

Computation example with R

```
n<-50
theta<-38.4
x<-rexp(n,1/theta)
mean(x)
alpha<-0.05
alpha1<-0.025
alpha2<-0.025
a<-2*n*mean(x)/qchisq(1-alpha1,2*n);a
b<-2*n*mean(x)/qchisq(alpha2,2*n);b
```

First strategy: pivotal function

Example: C.I for the parameters of a normal distribution

Theorem

Let $X_1, \dots, X_n$ be n i.i.d. r.v.'s from $\mathcal{N}(\mu, \sigma^2)$.

- (i) $\bar{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$.
- (ii) $(n-1)\hat{\sigma}_n^2/\sigma^2 \sim \chi_{n-1}^2$.
- (iii) $\bar{X}_n$ and $(n-1)\hat{\sigma}_n^2/\sigma^2$ are independent r.v.'s.

CI for the mean of a normal distribution

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where α_1 and α_2 are such that $\alpha_1 + \alpha_2 = \alpha$.

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$$a = \bar{X}_n - \frac{\sigma}{\sqrt{n}}\Phi^{-1}(1 - \alpha_1) \quad \text{and} \quad b = \bar{X}_n - \frac{\sigma}{\sqrt{n}}\Phi^{-1}(\alpha_2),$$

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- If $\alpha_1 = \alpha_2 = \alpha/2$, the interval is symmetric and the bounds are:

$$\bar{X}_n \pm \frac{\sigma}{\sqrt{n}}\Phi^{-1}(1 - \alpha/2)$$